EXPERIMENT NO. 1 -BASIC CALCULATOR

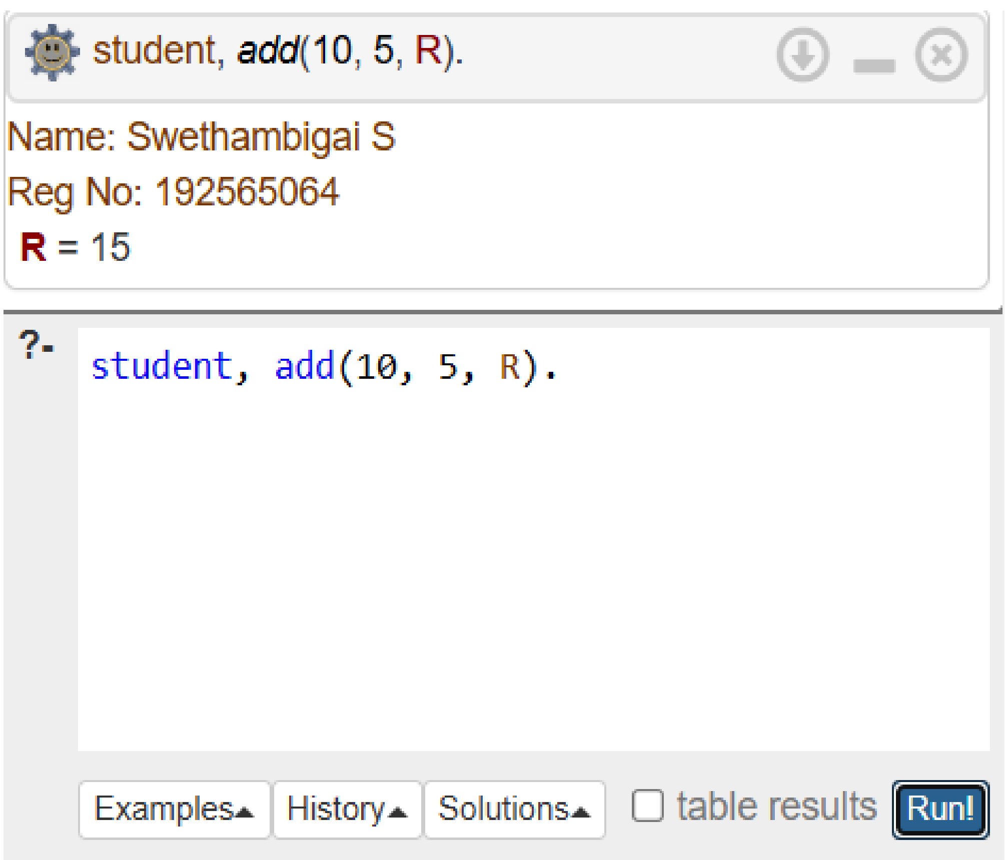
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

add(A, B, R) :-
    R is A + B.

subtract(A, B, R) :-
    R is A - B.

multiply(A, B, R) :-
    R is A * B.

divide(A, B, R) :-
    B =\= 0,
    R is A / B.
```

Query

```
student, add(10, 5, R).
```

Output



EXPERIMENT NO. 2-ALL COMPARISON

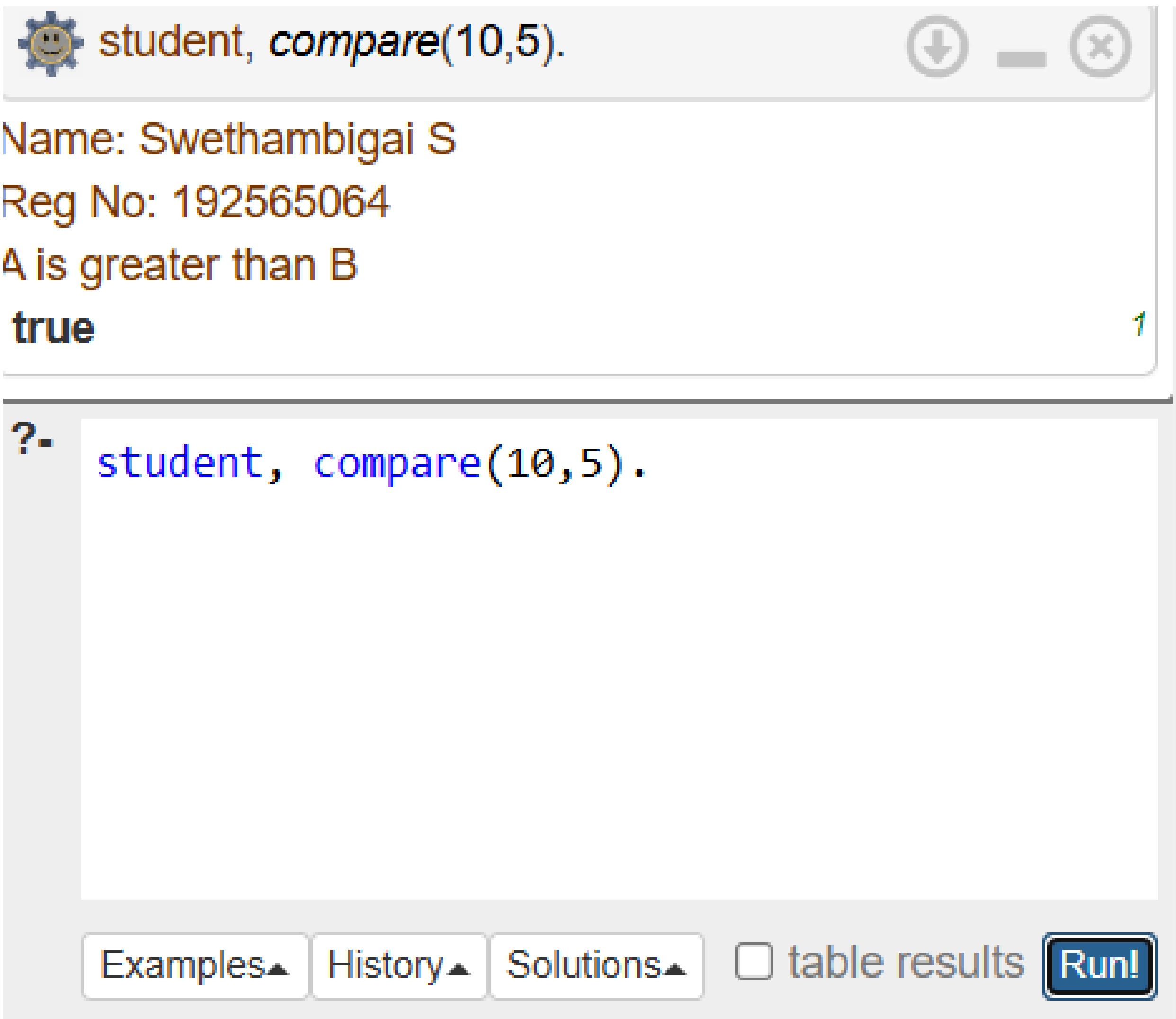
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

compare(A,B) :-
    A > B, write('A is greater than B').
```

Query

```
student, compare(10,5).
```

Output



EXPERIMENT NO. 3-VOTE ELIGIBILITY

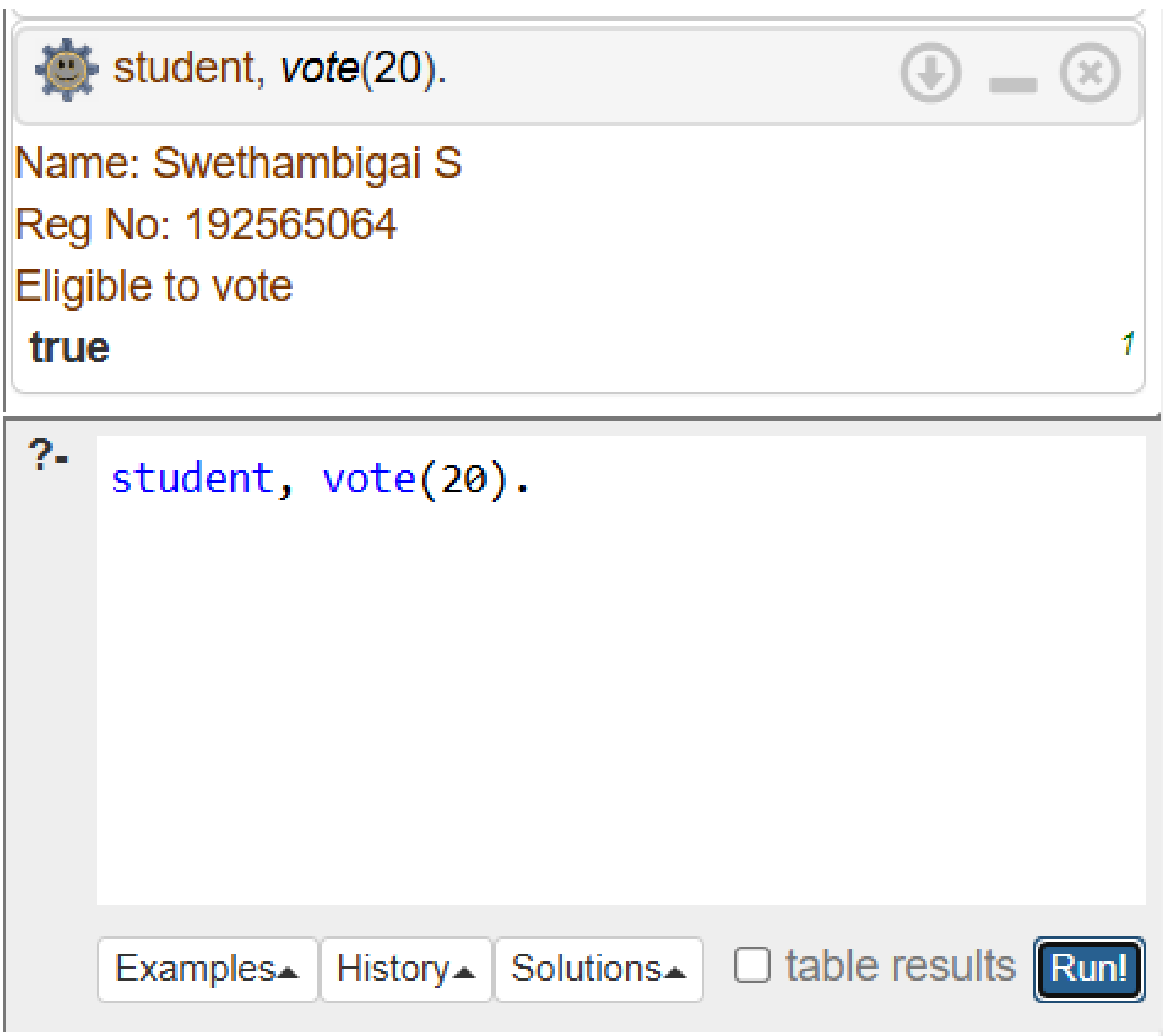
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

vote(Age) :-
    Age >= 18,
    write('Eligible to vote').
```

Query

```
student, vote(20).
```

Output



EXPERIMENT NO. 4-CELSIUS TO FAHRENHEIT

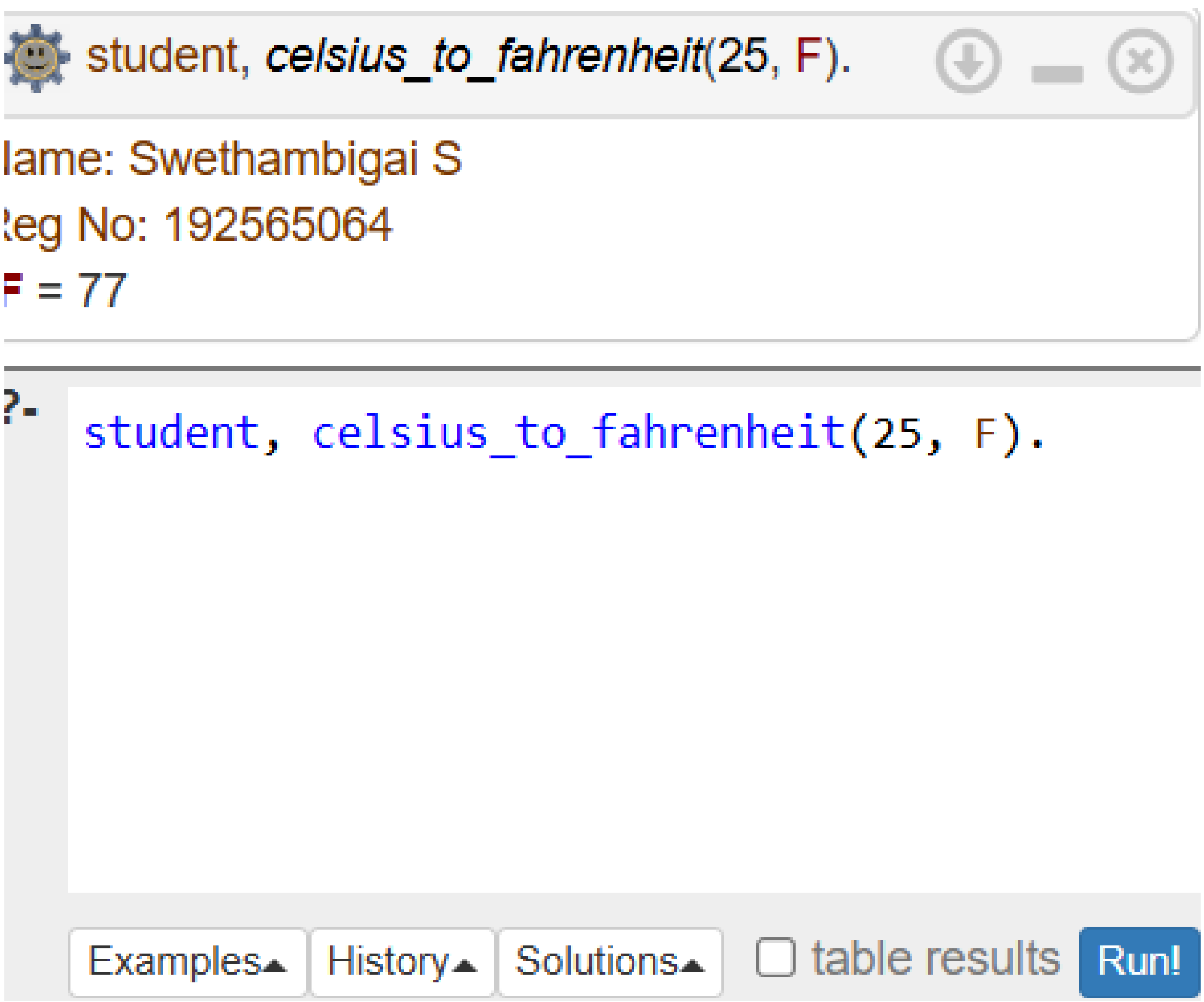
Prolog Code

```
student :-
    write('Name: Kaviya           '), nl,
    write('Reg No: 192565064'), nl.

celsius_to_fahrenheit(C, F) :-
    F is (C * 9 / 5) + 32.
```

Query

```
student, celsius_to_fahrenheit(25, F).
```

Output



EXPERIMENT NO. 5-PASS / FAIL – 3 SUBJECTS

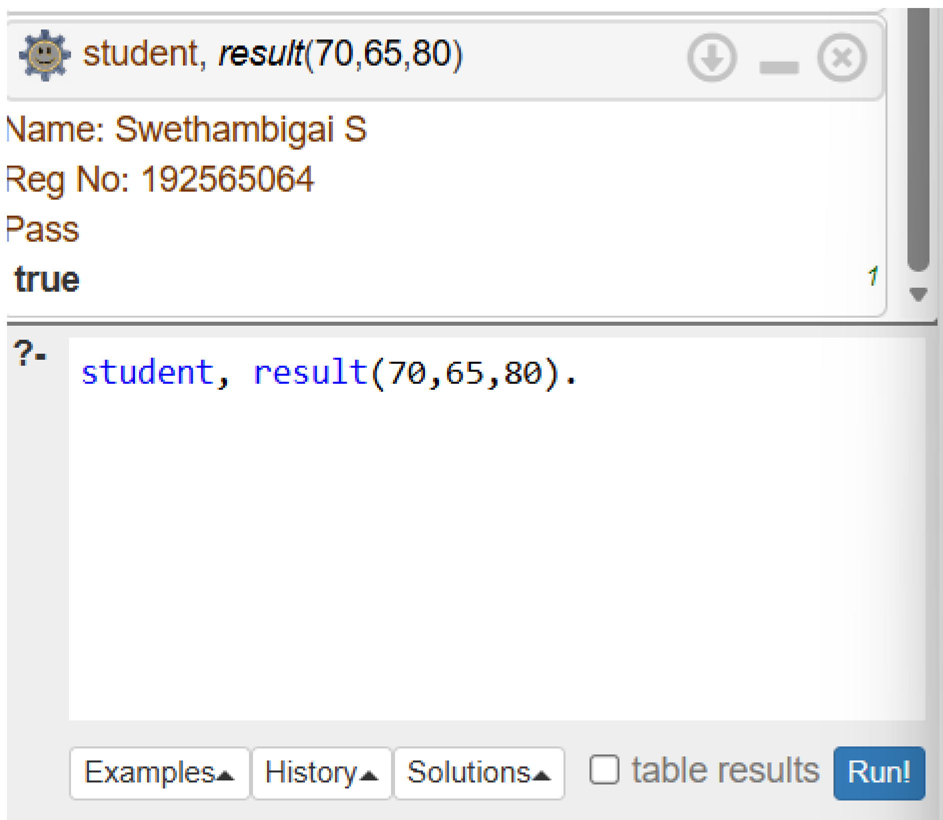
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

result(A,B,C) :-
    A >= 40, B >= 40, C >= 40,
    write('Pass').
```

Query

```
student, result(70,65,80).
```

Output



EXPERIMENT NO. 6- AVERAGE OF 5 NUMBERS

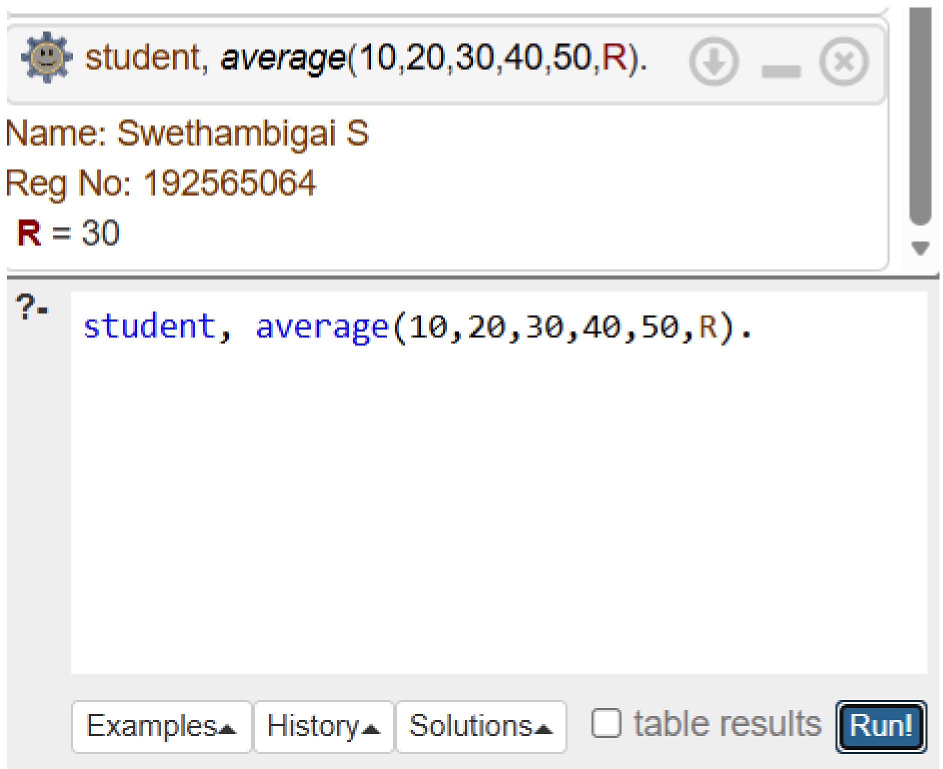
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

average(A,B,C,D,E,R) :-
    R is (A+B+C+D+E)/5.
```

Query

```
student, average(10,20,30,40,50,R).
```

Output



EXPERIMENT NO. 7-MULTIPLICATION TABLE

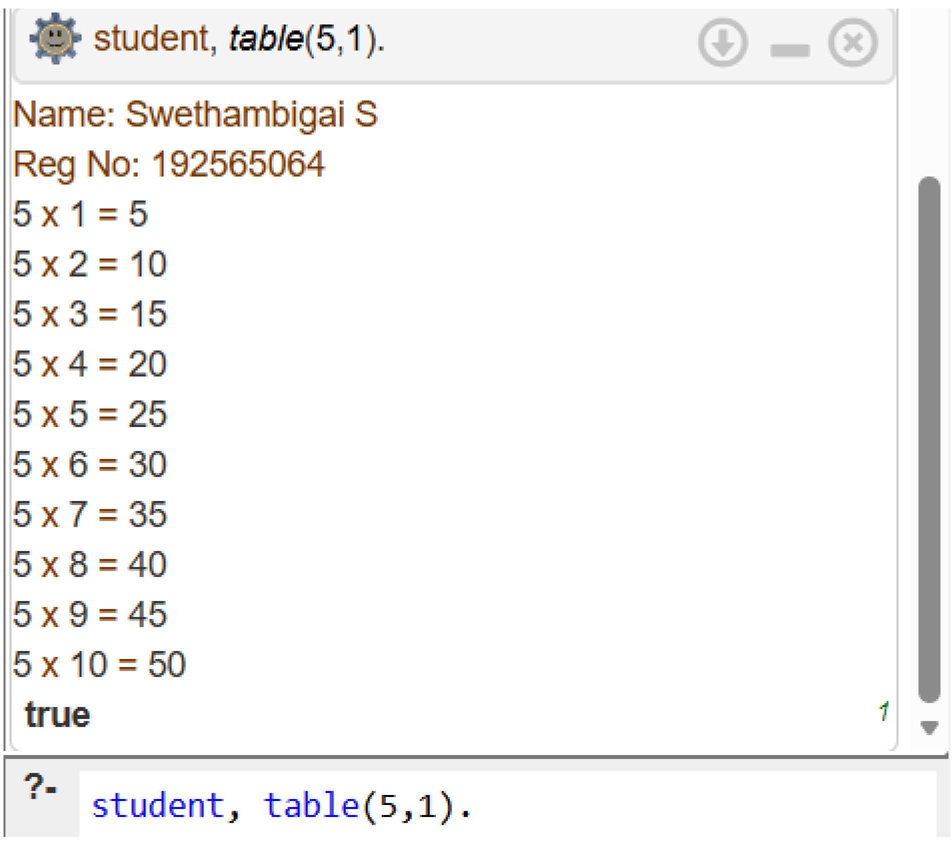
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

table(N,I) :-
    I =< 10,
    R is N*I,
    write(N), write(' x '), write(I), write(' = '), write(R), nl,
    I1 is I+1,
    table(N,I1).
table(_,11).
```

Query

```
student, table(5,1).
```

Output



EXPERIMENT NO. 8

STUDENT TEACHER SUBJECT RELATION

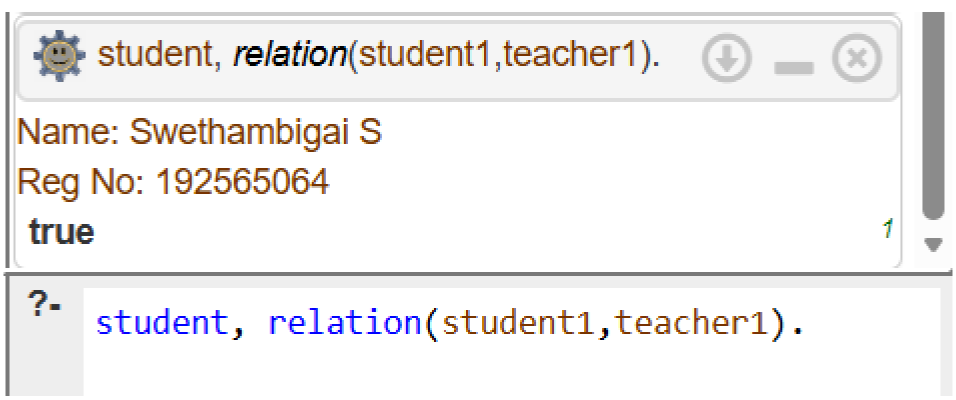
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

teaches(teacher1,prolog).
studies(student1,prolog).

relation(S,T) :-
    studies(S,prolog),
    teaches(T,prolog).
```

Query

```
student, relation(student1,teacher1).
```

Output



EXPERIMENT NO. 9

FAMILY TREE – ALL RELATIONS

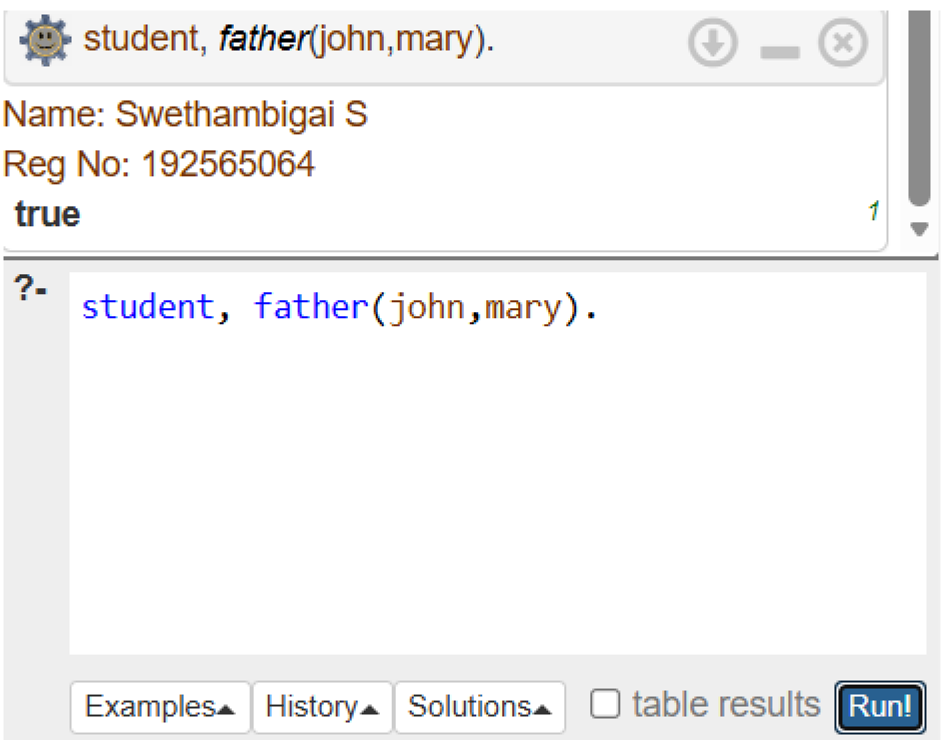
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

male(john).
female(mary).
parent(john,mary).

father(X,Y) :-
    male(X), parent(X,Y).

mother(X,Y) :-
    female(X), parent(X,Y).
```

Query

```
student, father(john,mary).
```

Output



EXPERIMENT NO. 10

DIETING SYSTEM BASED ON DISEASE

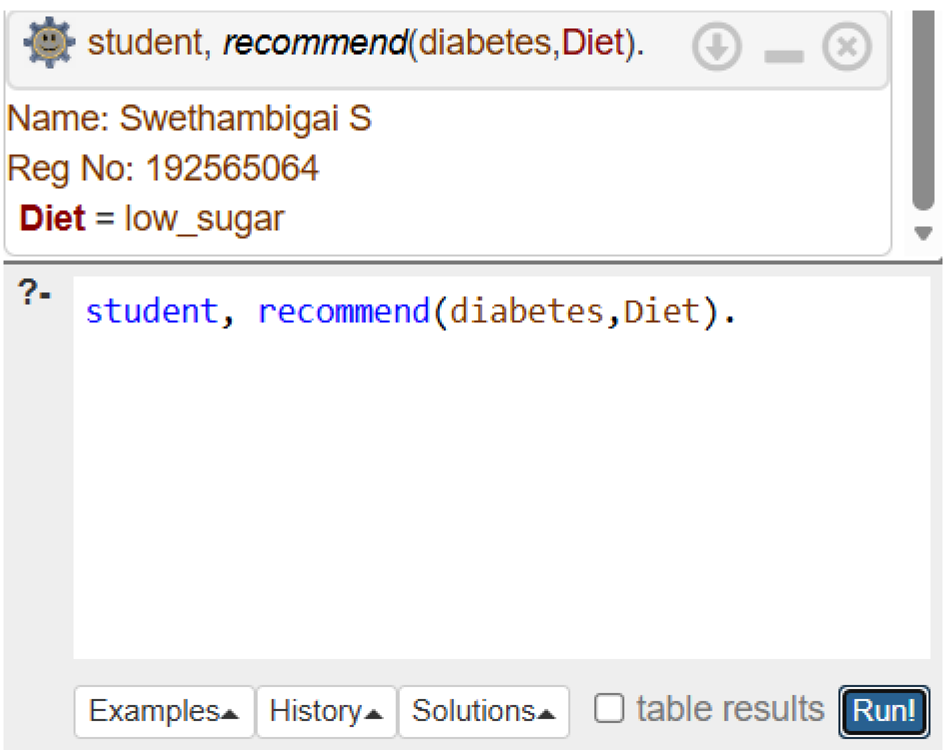
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

diet(diabetes,low_sugar).
diet(bp,low_salt).

recommend(D,Diet) :-
    diet(D,Diet).
```

Query

```
student, recommend(diabetes,Diet).
```

Output



EXPERIMENT NO. 11

FORWARD / BACKWARD CHAINING

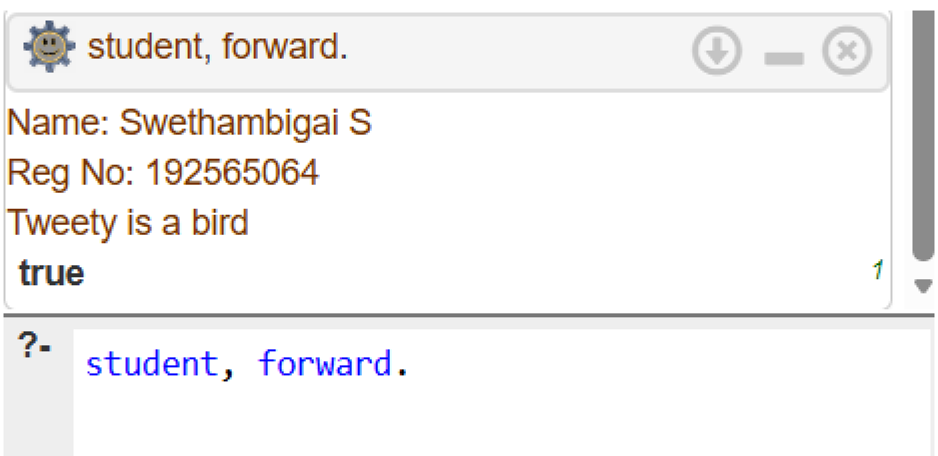
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

bird(tweety).
can_fly(tweety).

forward :-
    bird(tweety),
    write('Tweety is a bird').

backward :-
    can_fly(tweety),
    write('Tweety can fly').
```

Query

```
student, forward.
```

Output



EXPERIMENT NO. 12

MONKEY BANANA PROBLEM

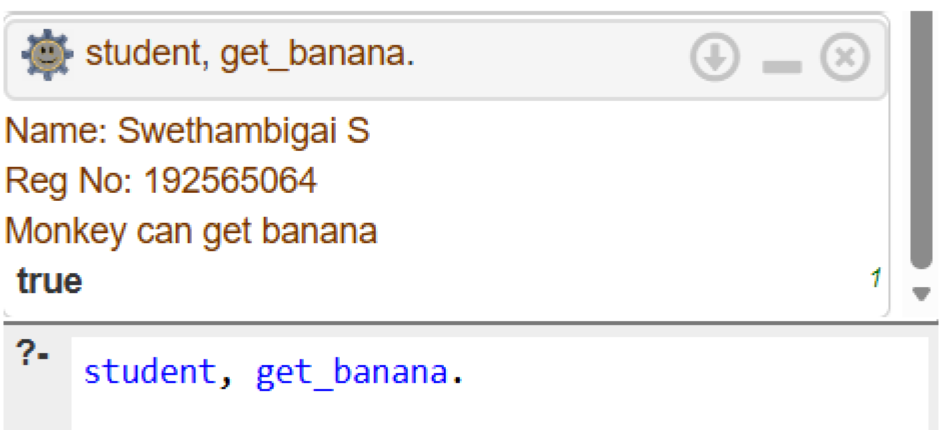
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

monkey_at(door).
box_at(window).
banana_at(ceiling).

get_banana :-
    monkey_at(door),
    box_at(window),
    banana_at(ceiling),
    write('Monkey can get banana').
```

Query

```
student, get_banana.
```

Output



EXPERIMENT NO. 13

FRUIT AND COLOR – BACKTRACKING

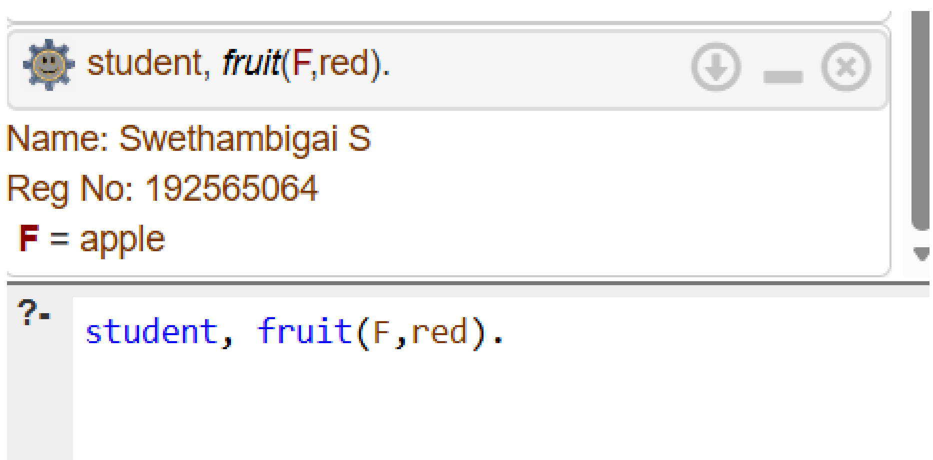
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

fruit(apple,red).
fruit(banana,yellow).
fruit(grape,green).
```

Query

```
student, fruit(F,red).
```

Output



EXPERIMENT NO. 14

BIRD FLY OR NOT

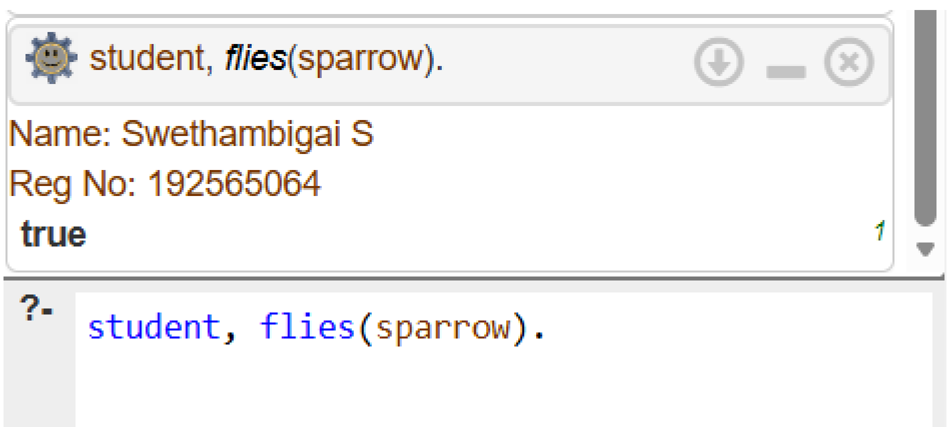
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

bird(sparrow).
bird(penguin).
can_fly(sparrow).

flies(X) :-
    bird(X), can_fly(X).
```

Query

```
student, flies(sparrow).
```

Output



EXPERIMENT NO. 15

GRADE FIND

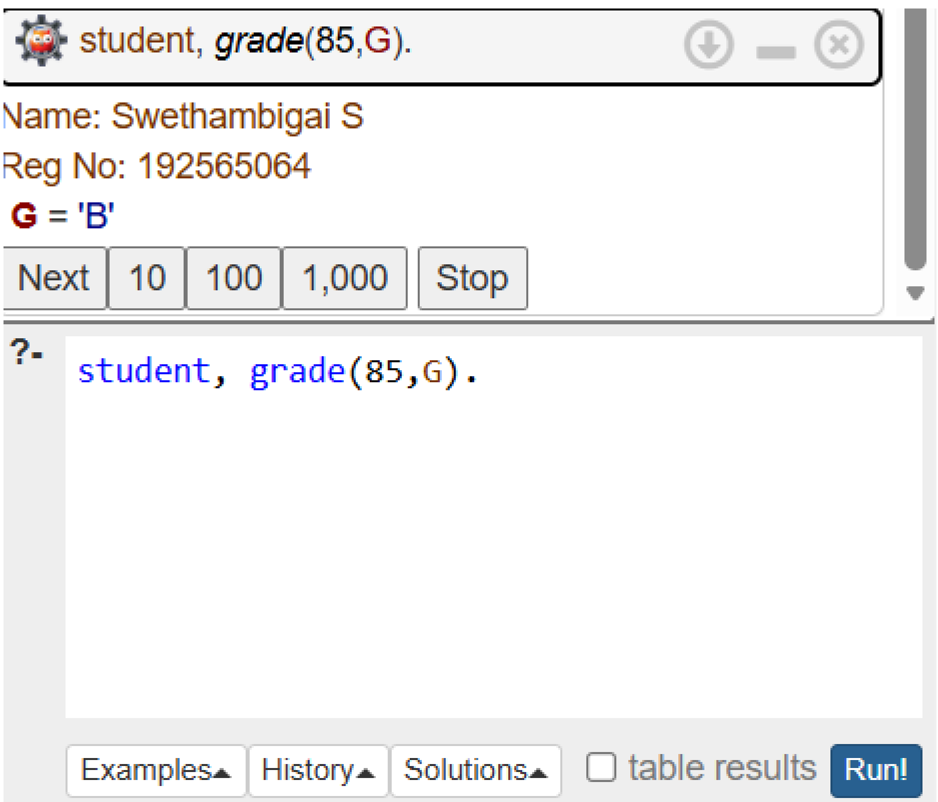
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

grade(M,A) :- M >= 90, A = 'A'.
grade(M,B) :- M >= 80, M < 90, B = 'B'.
grade(M,C) :- M >= 70, M < 80, C = 'C'.
grade(M,D) :- M >= 60, M < 70, D = 'D'.
grade(M,F) :- M < 60, F = 'F'.
```

Query

```
student, grade(85,G).
```

Output



EXPERIMENT NO. 16

EB BILL PROGRAM

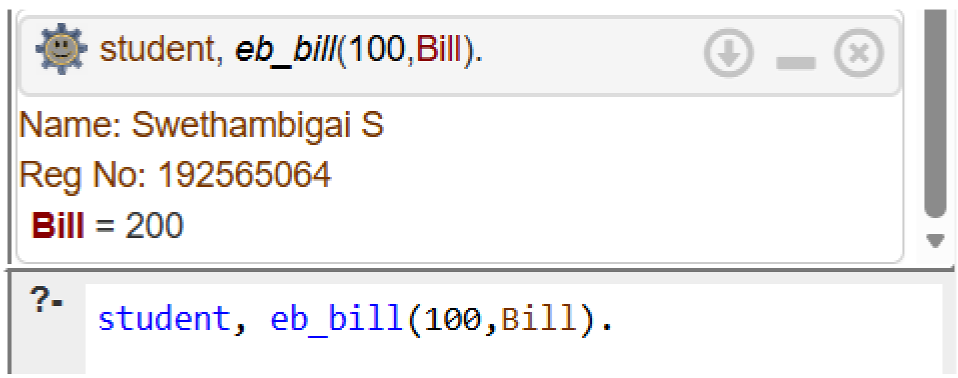
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

eb_bill(Units,Bill) :-
    Bill is Units * 2.
```

Query

```
student, eb_bill(100,Bill).
```

Output



EXPERIMENT NO. 17

STUDENT DATABASE

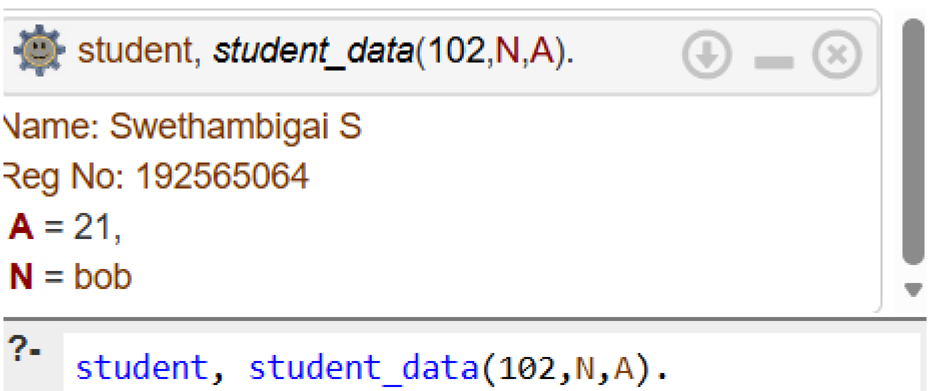
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

student_data(101,alice,20).
student_data(102,bob,21).
student_data(103,charlie,19).
```

Query

```
student, student_data(102,N,A).
```

Output



EXPERIMENT NO. 18

PRINT 1 TO 10

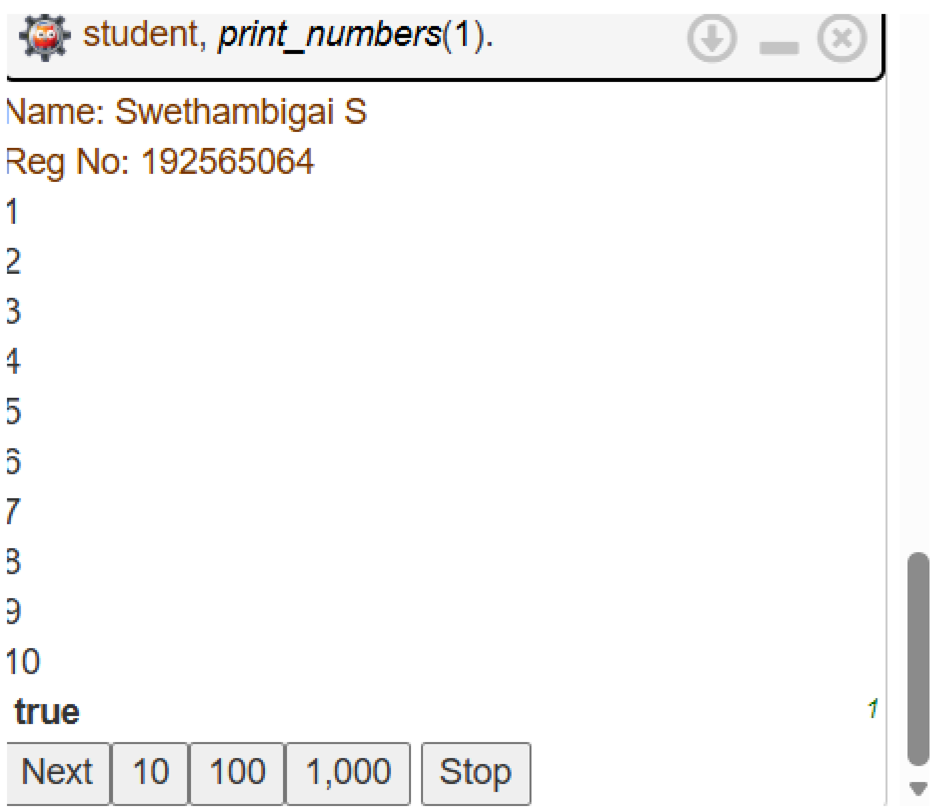
Prolog Code

```
student :-
    write('Name: Kaviya            '), nl,
    write('Reg No: 192565064'), nl.

print_numbers(11).
print_numbers(N) :-
    write(N), nl,
    N1 is N+1,
    print_numbers(N1).
```

Query

```
student, print_numbers(1).
```

Output



EXPERIMENT NO. 19

N QUEENS PROBLEM

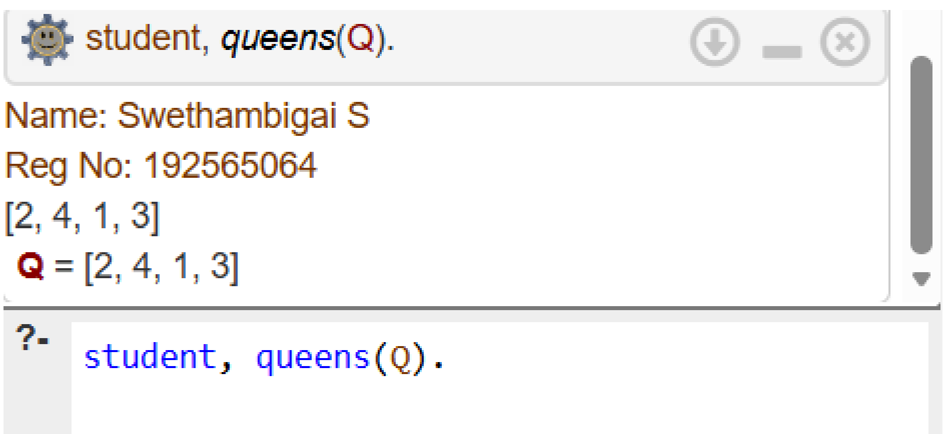
Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

queens(Q) :-
    Q = [2,4,1,3],
    write(Q).
```

Query

```
student, queens(Q).
```

Output



EXPERIMENT NO. 20

REVERSE A NUMBER

Prolog Code

```
student :-
    write('Name: Kaviya          '), nl,
    write('Reg No: 192565064'), nl.

reverse_number(N,R) :-
    reverse_number(N,0,R).

reverse_number(0,R,R).
reverse_number(N,A,R) :-
    N > 0,
    D is N mod 10,
    N1 is N // 10,
    A1 is A*10+D,
    reverse_number(N1,A1,R).
```

Query

```
student, reverse_number(1234,R).
```

Output